Formal Specification of the Expansion-Compression Cerebellar Manifold (ECCM)

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Abstract

This document formally defines the mathematical and biophysical architecture of the Expansion-Compression Cerebellar Manifold (ECCM). Based on the recent topological diagnostics and the causal Recursive Least Squares (RLS) fading-window tournament, the model implements a three-stage topological pipeline: (1) Temporal Sensory Expansion, (2) Symplectic Dimensional Compression, and (3) Asymmetric Analytic Plasticity. This formalization corresponds directly to the diagnostic which achieved the theoretical upper-bound lower deviance of 152.31 at $N_{comp} = 20$.

1 Stage 1: The Afferent Temporal State Vector (Mossy Fibers)

The external state of the environment is mapped into the cerebellar manifold via the Mossy Fiber (MF) afferents. To enforce strict biophysical compliance, the observable state history is tracked at every trial t using only isolated sensorimotor parameters.

Let the raw observable state vector at trial t be defined as $S(t) \in \mathbb{R}^6$:

$$S(t) = \left[c_{t-1}, o_{t-1}, \tilde{RT}_{t-1}, R_1^{cf}, R_2^{cf}, \bar{c}_{t-1}\right]^T \tag{1}$$

where c_{t-1} is the previous choice, o_{t-1} is the previous physical outcome, $\tilde{RT}_{t-1}$ is the standardized previous reaction time, R_i^{cf} represents the counterfactual or actual reward of option i, and $\bar{c}_{t-1}$ is the unchosen option.

The Mossy Fibers (N_{MF}) construct a temporal reservoir by sampling delayed representations of this state space via a non-linear sigmoid basis. For each Mossy Fiber $j \in \{1, \ldots, N_{MF}\}$:

$$u_j(t) = \begin{cases} \sigma\left(\beta_j \cdot S_{c_j}(t)\right) & \text{if } d_j = 0 \\ \sigma\left(\beta_j \cdot \left(S_{c_j}(t) - S_{c_j}(t - d_j)\right)\right) & \text{if } d_j > 0 \end{cases} \tag{2}$$

where $\sigma(x) = (1 + \exp(-x))^{-1}$, $c_j \sim \mathcal{U}\{1, 6\}$ is the sampled state dimension, $d_j \sim \mathcal{P}(\lambda_d)$ is the sampled temporal delay (acting as a delay-line), and $\beta_j \sim \mathcal{L}(\mu_\beta, \sigma_\beta)$ is the log-normal spatial gain.

2 Stage 2: The High-Dimensional Expansion (Granular Layer)

The Mossy Fiber signals undergo massive geometric expansion into the Granular Layer (N_{GC}). The Granule cells integrate spatial fan-in from the Mossy Fibers with local temporal persistence.

Each Granule cell i randomly samples $k = 4$ Mossy Fibers. The instantaneous spatial drive is:

$$\text{Drive}_i(t) = \sum_{m=1}^{4} W_{i,m}^{MF\to GC} \cdot u_{\text{map}(i,m)}(t) \tag{3}$$

where $W_{i,m}^{MF\to GC} \in \{-1, 1\}$ are sparse binary projection weights.

To generate a fading-memory temporal topology, each Granule cell maintains a dynamic trace governed by a continuous biophysical decay. The decay factor $\gamma_i(t)$ depends on the inter-trial delay Δt and the cell's intrinsic time constant τ_i:

$$\gamma_i(t) = \rho_{base} + (1 - \rho_{base}) \exp\left(-\frac{\Delta t}{\tau_i}\right) \tag{4}$$

The fully expanded Granular state vector $z_{GC}(t) \in \mathbb{R}^{N_{GC}}$ is thus defined recursively:

$$z_{GC,i}(t) = \text{Drive}_i(t) + \gamma_i(t) \cdot z_{GC,i}(t-1) \tag{5}$$

3 Stage 3: The Topological Compression Matrix (Golgi/Phase 2)

To prevent algebraic collapse and isolate the linearly separable dimensions of the expanded manifold, the Granular space is projected through a fixed, random compression matrix $W_{comp} \in \mathbb{R}^{N_{comp} \times N_{GC}}$ into a compressed bottleneck $z_{comp}(t) \in \mathbb{R}^{N_{comp}}$, followed by a non-linear thresholding function (ReLU).

For each compression dimension $j \in \{1, \ldots, N_{comp}\}$:

$$z_{comp,j}(t) = \max\left(0, \sum_{i=1}^{N_{GC}} W_{comp,j,i} \cdot z_{GC,i}(t)\right) \tag{6}$$

where $W_{comp,j,i} \sim \mathcal{N}\left(0, \frac{1}{\sqrt{N_{GC}}}\right)$. This step represents the critical mathematical mechanism that stabilized the manifold during the $N_{comp} \in \{10, 20, 40\}$ diagnostics.

4 Stage 4: Asymmetric Readout and the Causal Fading Window

The cerebellar policy readout translates the compressed manifold into a value/motor policy $V_{PC}(t)$ via Purkinje cell weights $w_a(t)$. For a chosen action $a(t)$:

$$V_{PC}(t) = \sum_{j=1}^{N_{comp}} w_{a(t),j}(t) \cdot z_{comp,j}(t) \tag{7}$$

The Inferior Olive (IO) error is derived from the environmental outcome target $T(t) = \pm 1$:

$$\delta_{IO}(t) = T(t) - V_{PC}(t) \tag{8}$$

4.1 Analytic Causal Plasticity (Recursive Least Squares)

To evaluate the absolute mathematical upper bound of the manifold's separability, the Purkinje weights are updated using the exact, causal Recursive Least Squares (RLS) fading-window solution, mathematically identical to the optimal theoretical performance of the continuous asymmetric L_2 Ridge Regularization over deep time.

Let $P(t) \in \mathbb{R}^{N_{comp} \times N_{comp}}$ be the inverse covariance matrix, and $\gamma_{RLS} \in [0.5, 1.0]$ be the fading memory factor. The Kalman gain $K(t) \in \mathbb{R}^{N_{comp}}$ is computed as:

$$K(t) = \frac{P(t-1) z_{comp}(t)}{\gamma_{RLS} + z_{comp}(t)^T P(t-1) z_{comp}(t)} \tag{9}$$

The Purkinje weights $w_{a(t)}$ and the inverse covariance matrix are updated strictly causally:

$$w_{a(t)}(t) = w_{a(t)}(t-1) + K(t) \cdot \delta_{IO}(t) \tag{10}$$

$$P(t) = \frac{1}{\gamma_{RLS}} \Big(P(t-1) - K(t) z_{comp}(t)^T P(t-1) \Big) \tag{11}$$